ALA-2
Python Deep Dive: Interview Intensive
Topic Name: Tuple

1. What is a Tuple in Python?
Answer:

- A tuple is a built-in data type in Python that is used to store multiple items in a single variable.

- It is ordered
- It is immutable (cannot be changed)
- Allows duplicate values

Example:
t = (1, 2, 3, 4)
Explanation:
Once a tuple is created, you cannot modify, add, or remove elements.

2. Difference between List and Tuple?
Answer:

Feature	List	Tuple
Mutability	Mutable	Immutable
Syntax	[ ]	( )
Performance	Slower	Faster
Memory Usage	More	Less
Use Case	Dynamic data	Fixed data

3.Why are Tuples Immutable?
Answer:
Tuples are immutable to:

- Ensure data safety
- Improve performance
- Allow usage as dictionary keys

Example:
t = (1, 2, 3)
t[0] = 10 # Error

4. How to Create a Tuple?
Answer:

Different ways:
```
t1 = (1, 2, 3)
t2 = 1, 2, 3
t3 = tuple([1, 2, 3])
```
Single element tuple:
```
t = (5,)   # comma is required
```

5. Tuple Packing and Unpacking?
Answer:
Packing:
```
t = (1, 2, 3)
```

Unpacking:
```
a, b, c = t
print(a, b, c)
```

Advanced Unpacking:
```
a, *b = (1, 2, 3, 4)
print(a)  # 1
print(b)  # [2, 3, 4]
```

6. Can Tuples Contain Mutable Objects?
Answer:
Yes, tuples can contain mutable objects like lists.
```
t = (1, [2, 3])
t[1][0] = 100
print(t)
```

Explanation:
- Tuple itself is immutable
- But mutable elements inside it can change

7. Tuple Methods?
Answer:
Tuples have only 2 built-in methods:

1. count() – Counts occurrences
2. index() – Finds position

```
t = (1, 2, 2, 3)
print(t.count(2))   # 2
print(t.index(3))   # 3
```

8. Tuple vs Set?
Answer:

Feature	Tuple	Set
Order	Ordered	Unordered
Mutability	Immutable	Mutable
Duplicates	Allowed	Not allowed
Indexing	Yes	No

9. Why Tuples are Faster than Lists?
Answer:

- Fixed size → No dynamic resizing
- Less memory usage
- Optimized internal structure

10. When Should You Use Tuples?
Answer:
Use tuples when:
✓ Data should not change
✓ Faster performance required
✓ Used as dictionary keys
✓ Represent fixed collections (e.g., coordinates)

Example:
location = (20.5937, 78.9629)